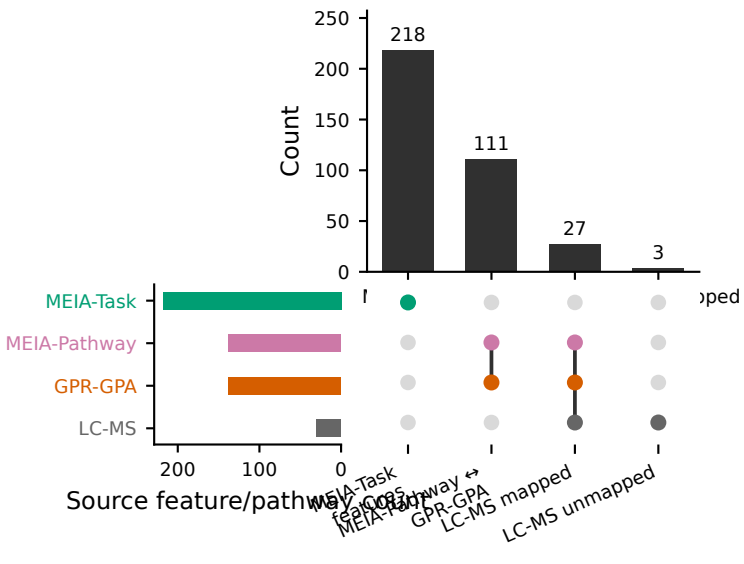
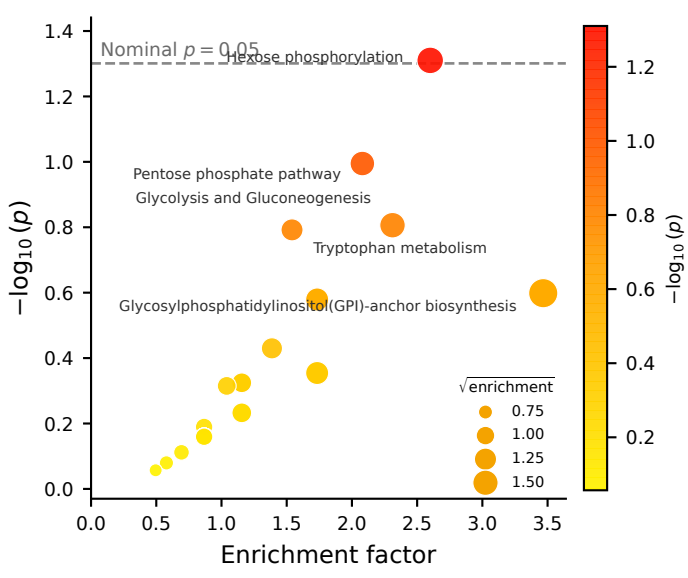
## SA Metastatic-status four-set metabolic overlap

LC-MS: 34 pathways → 30 harmonized
(27 mapped + 3 unmapped); MEIA: 218 tasks + 138 pathways.



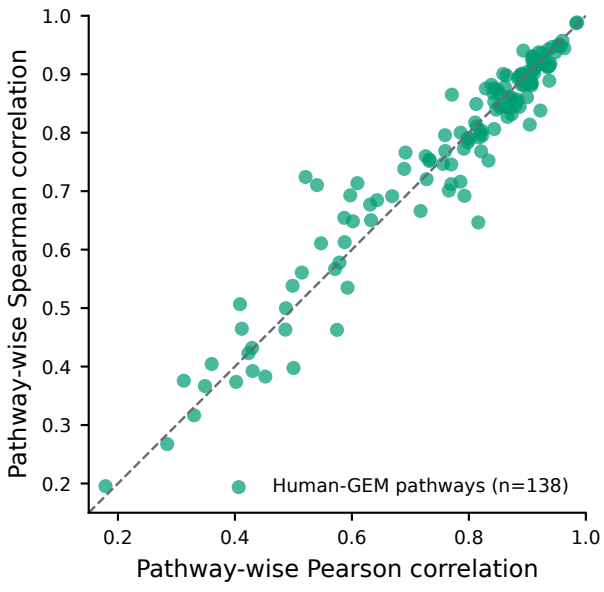

## SB Metastatic-status pathway enrichment

34 candidates shared with SA; top five labeled.
0 pass pathway BH-FDR < 0.05.




## SJ GPR-GPA–scKNIFE concordance

138 Human-GEM pathways; each correlated across 69,390 cells




## S1 Patient robustness of MEIA–scKNIFE concordance

Dots are patient medians; intervals are within-patient cell IQRs across 356 entities.



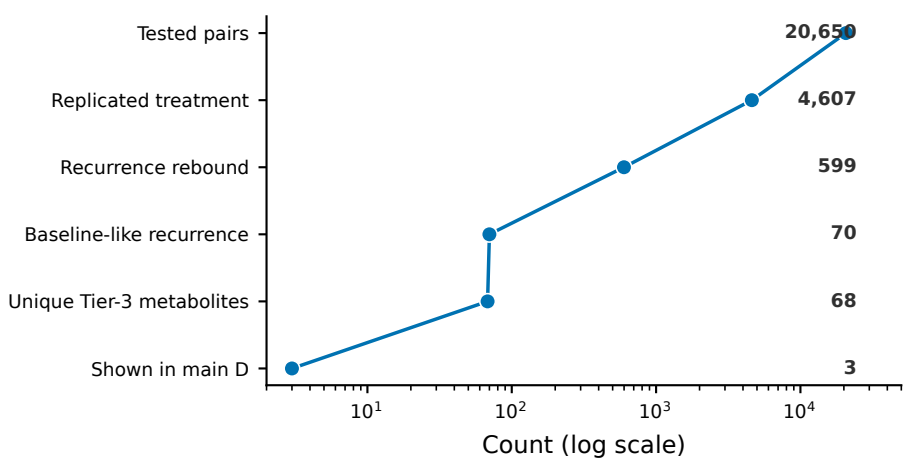




Descriptive technical concordance; no inferential p-values.

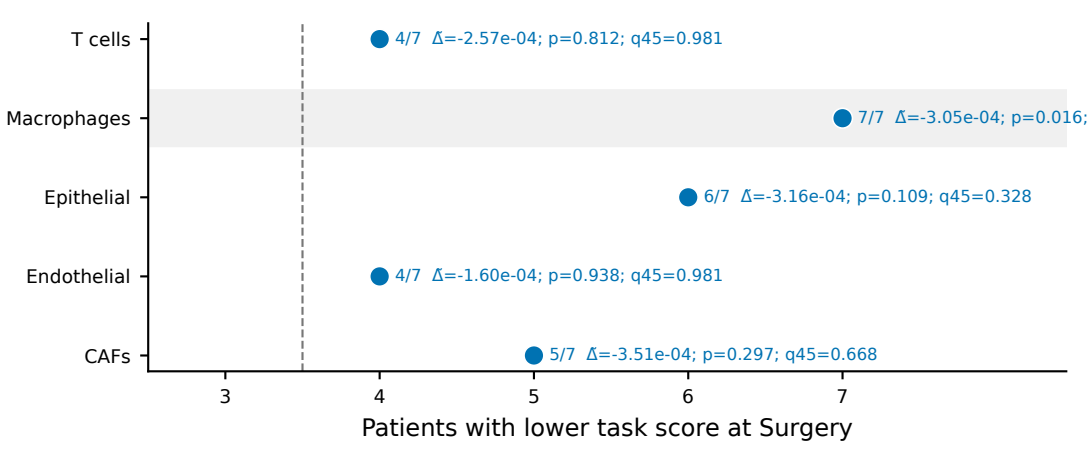
## SFUNNEL Robust inferred-metabolite filtering cascade

Metabolite–cell-type pairs are distinguished from unique inferred metabolite nodes.



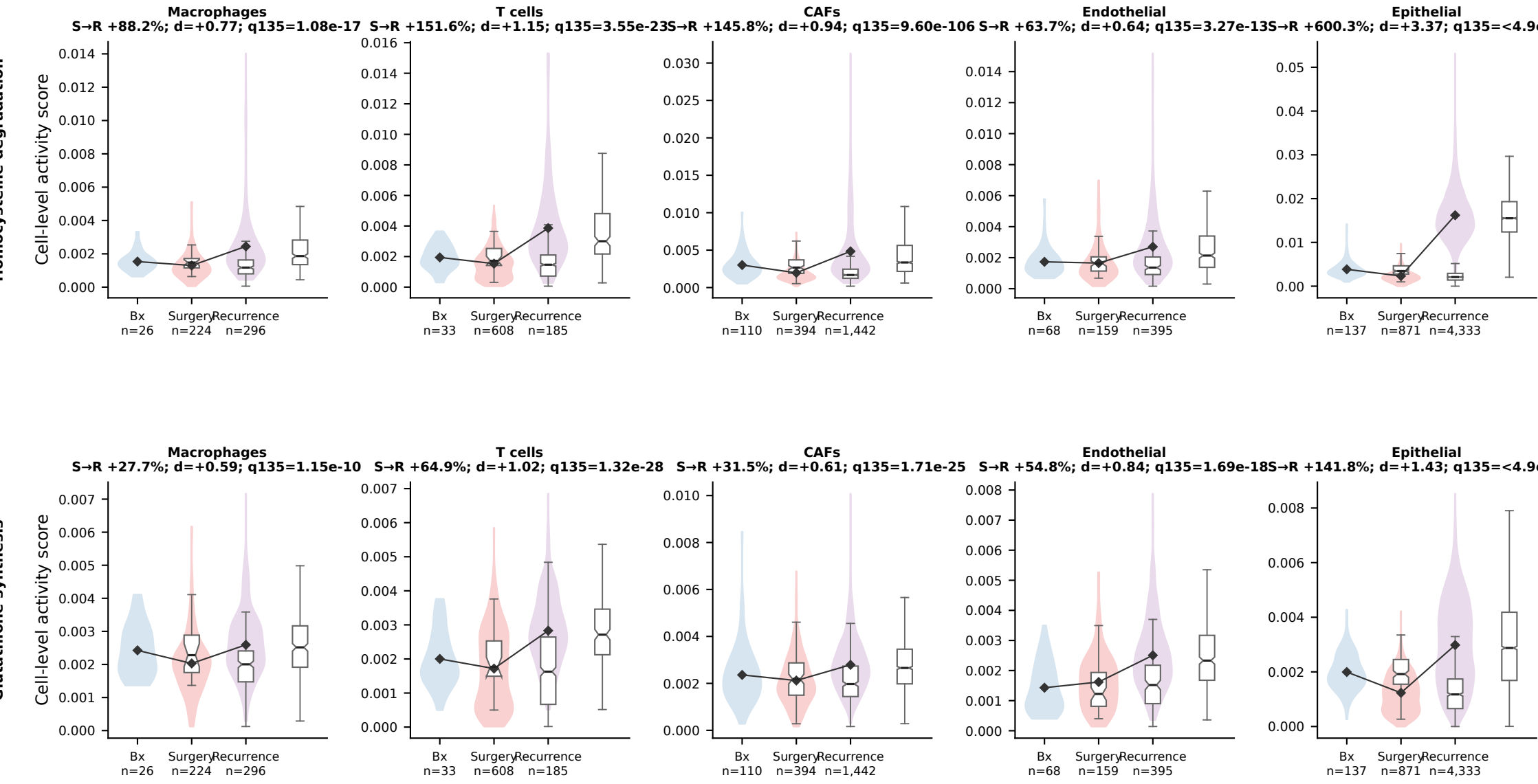

## SDIRECTION Homocysteine degradation across cell types

Exact paired Wilcoxon tests across seven patient medians; BH across the original 45 tests.




Macrophages: 7/7 decrease, nominal p=0.0156, q45=0.281; directionally consistent but not FDR-significant.

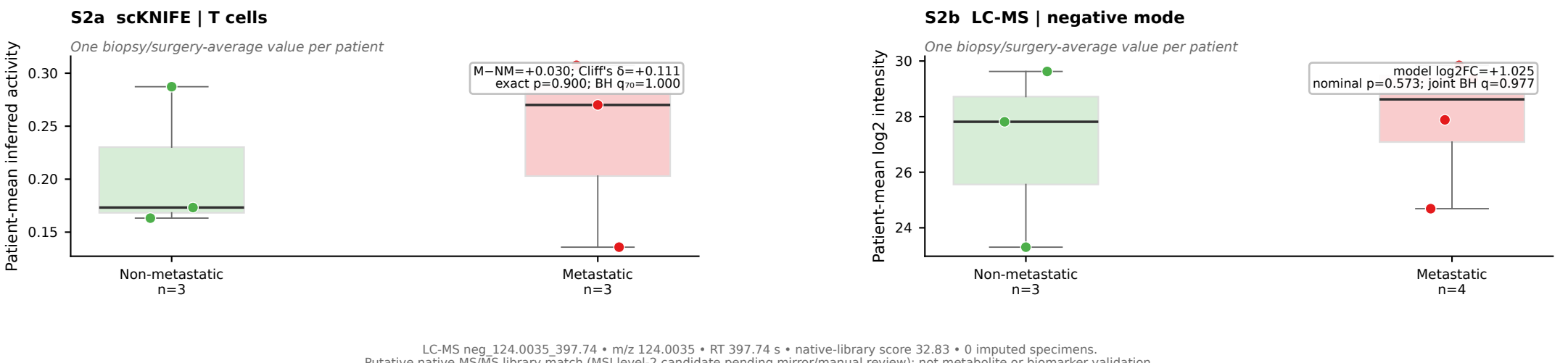
## SU3 P3 homocysteine-degradation and glutathione-synthesis trajectories




Violin width=density; white box=IQR; line=median; whiskers=1.5×IQR; diamond/line=mean; box outliers hidden.
Welch tests; BH across original 135 tests. One patient; cells are pseudoreplicates.

## S2 2-Aminoethylphosphonate status candidate

Directionally concordant but statistically unsupported across scKNIFE and LC-MS.




LC-MS neg_124.0035_397.74 • m/z 124.0035 • RT 397.74 s • native-library score 32.83 • 0 imputed specimens.
Putative native MS/MS library match (MSI level-2 candidate pending mirror/manual review); not metabolite or biomarker validation.